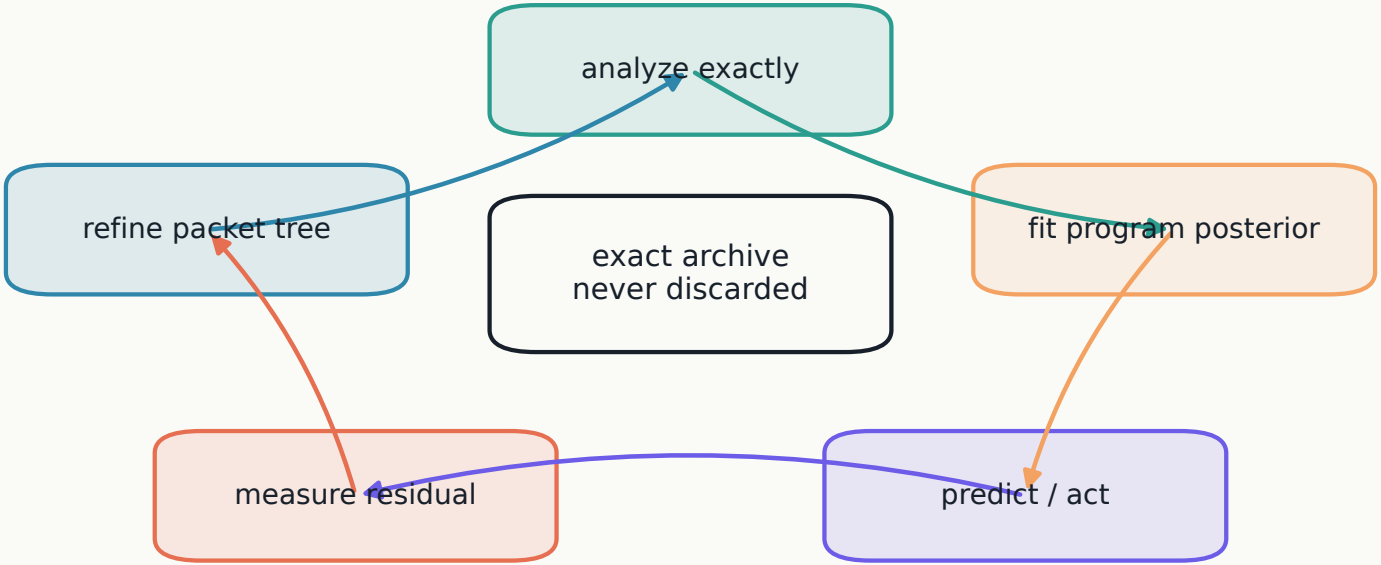Residual-driven learning cycle